

Best Programming Practice

1. All values as variables including Fixed, User Inputs, and Results
2. Proper naming conventions for all variables

```
String name = "Eric";  
double height = input.nextDouble();  
double totalDistance = distanceFromToVia + distanceViaToFinalCity;
```

3. Proper Program Name and Class Name
4. Follow proper indentation
5. Give comments for every step or logical block like a variable declaration or conditional and loop blocks

1. **Sample Program 1** - Create a program to check if 3 values are internal angles of a triangle.

IMP => Follow Good Programming Practice demonstrated below in all Practice Programs

Hint =>

- a. Get integer input for 3 variables named x, y, and z.
- b. Find the sum of x, y, and z.
- c. If the sum is equal to 180, print "The given angles are internal angles of a triangle" else print They are not

Java

```
// Creating Class with name TriangleChecker indicating the purpose is to  
// check if the internal angles add to 180  
import java.util.Scanner;  
  
class TriangleChecker {  
    public static void main(String[] args) {  
        // Create a Scanner Object  
        Scanner input = new Scanner(System.in);  
  
        // Get 3 input values for angles  
        int x = input.nextInt();  
        int y = input.nextInt();  
        int z = input.nextInt();  
  
        // Find the sum of all angles  
        int sumOfAngles = x + y + z;  
  
        // Check if sum is equal to 180 and print either true or false  
        System.out.println("The given angles " +x+ ", " +y+ ", " + z +  
            " add to " + sumOfAngles);
```

```

if (sumOfAngles == 180) {
    System.out.println("The given angles are internal angles of a " +
        "Triangle");
} else {
    System.out.println("The given angles are not internal angles " +
        "of a Triangle");
}

// Closing the Scanner Stream
input.close();
}
}

```

2. **Sample Program 2** - Create a program to find the sum of all the digits of a number given by a user.

Hint =>

- Get an integer input for the number variable.
- Create an integer variable sum with an initial value of 0.
- Create a while loop to access each digit of the number.
- Inside the loop, add each digit of the number to the sum.
- Finally, print the sum outside the loop

Java

```

// Create SumOfDigit Class to compute the sum of all digits of a number
import java.util.Scanner;

class SumOfDigits {

    public static void main(String[] args) {
        // Create a Scanner Object
        Scanner input = new Scanner(System.in);

        // Get input value for number
        int origNumber = input.nextInt();

        // Define variable number and sum initialized to zero
        int number = origNumber;
        int sum = 0;
    }
}

```

```
// Run while loop to access each digit of number
while (number != 0) {
    // Use number % 10 to find each digit of number from last
    int digit = number % 10;

    // add each digit to sum
    sum += digit;

    // Remove last digit from number essentially get the quotient
    number = number / 10;
}

// Print the sum and close the Scanner Stream
System.out.println("The sum of digit of number:" +origNumber+ " = " +
                    sum);
input.close();
}
}
```

Level 3 Practice Programs

1. Create a program to check if a number is Armstrong or not. Use the hints to show the steps clearly in the code

Hint =>

- a. Armstrong Number is a number whose Sum of cubes of each digit results in the original number e.g. $153 = 1^3 + 5^3 + 3^3$
- b. Get an integer input and store it in the number variable define sum variable, initialize it to zero and originalNumber variable, and assign it to the input number variable
- c. Use the **while** loop till the originalNumber is not equal to zero
- d. In the **while** loop find the remainder number by using the modulus operator as in **number % 10**. Find the cube of the number and add it to the **sum** variable
- e. Again in while loop find the quotient of the number and assign it to the original number using number / 10 expression. This removes the last digit of the original number.
- f. Finally check if the number and the sum are the same, if same its an Armstrong number else not. So display accordingly

```
import java.util.Scanner;
public class ArmstrongNumber {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Get integer input and store it in the number variable
        System.out.print("Enter an integer: ");
        int number = input.nextInt();

        // Initialize sum variable to zero and originalNumber variable
        to the input number
        int sum = 0;
        int originalNumber = number;

        // Use while loop till originalNumber is not equal to zero
        while (originalNumber != 0) {

            // Find the remainder (last digit) using the modulus
            operator
            int remainder = originalNumber % 10;
```

```

        // Find the cube of the remainder and add it to the sum
        variable
        sum += remainder * remainder * remainder;

        // Find the quotient of the number and assign it to
        originalNumber (remove the last digit)
        originalNumber /= 10;
    }

    // Check if the original number and the sum are the same
    if (sum == number) {
        System.out.println(number + " is an Armstrong number.");
    }
    else {
        System.out.println(number + " is not an Armstrong number.");
    }
    // Close the scanner
    input.close();
}
}

```

2. Create a program to count the number of digits in an integer.

Hint =>

- Get an integer input for the number variable.
- Create an integer variable count with value 0.
- Use a loop to iterate until number is not equal to 0.
- Remove the last digit from number in each iteration
- Increase count by 1 in each iteration.
- Finally display the count to show the number of digits

```

import java.util.Scanner;
public class DigitCounter {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Get an integer input for the number variable
        System.out.print("Enter an integer: ");
        int number = input.nextInt();
    }
}

```

```
// Initialize the count variable to 0
int count = 0;

// Use a loop to iterate until number is not equal to 0
while (number != 0) {

    // Remove the last digit from the number
    number /= 10;
    // Increase count by 1 in each iteration
    count++;
}

// Display the count to show the number of digits
System.out.println("The number of digits is: " + count);

// Close the scanner
input.close();
}
}
```

3. Create a program to check if a number taken from the user is a Harshad Number.

Hint =>

- A Harshad number is an integer which is divisible by the sum of its digits.
For example, 21 which is perfectly divided by 3 (sum of digits: 2 + 1).
- Get an integer input for the number variable.
- Create an integer variable sum with initial value 0.
- Create a while loop to access each digit of the number.
- Inside the loop, add each digit of the number to sum.
- Check if the number is perfectly divisible by the sum.
- If the number is divisible by the sum, print Harshad Number. Otherwise, print Not a Harshad Number.

```
import java.util.Scanner;
public class HarshadNumber {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
```

```

Scanner input = new Scanner(System.in);

// Get integer input for the number variable
System.out.print("Enter an integer: ");
int number = input.nextInt();

// Initialize the sum variable to 0
int sum = 0;
int originalNumber = number;

// Create a while loop to access each digit of the number
while (originalNumber != 0) {

    // Find the remainder (last digit) using the modulus
operator    int remainder = originalNumber % 10;

    // Add the digit to the sum variable
    sum += remainder;

    // Remove the last digit from the original number
    originalNumber /= 10;
}

// Check if the number is perfectly divisible by the sum
if (number % sum == 0) {
    System.out.println(number + " is a Harshad number.");
}
else {
    System.out.println(number + " is not a Harshad number.");
}
// Close the scanner
input.close();
}
}

```

4. Create a program to check if a number is an Abundant Number.

Hint =>

- a. An abundant number is an integer in which the sum of all the divisors of the number is greater than the number itself. For example,

Divisor of 12: 1, 2, 3, 4, 6

Sum of divisor: $1 + 2 + 3 + 4 + 6 = 16 > 12$

- b. Get an integer input for the number variable.
- c. Create an integer variable sum with initial value 0.
- d. Run a for loop from $i = 1$ to $i < \text{number}$.
- e. Inside the loop, check if number is divisible by i .
- f. If true, add i to sum.
- g. Outside the loop Check if sum is greater than number.
- h. If the sum is greater than the number, print Abundant Number. Otherwise, print Not an Abundant Number.

```
import java.util.Scanner;
public class AbundantNumber {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Get an integer input for the number variable
        System.out.print("Enter an integer: ");
        int number = input.nextInt();

        // Initialize the sum variable to 0
        int sum = 0;

        // Run a for loop from i = 1 to i < number
        for (int i = 1; i < number; i++) {

            // Check if number is divisible by i
            if (number % i == 0) {

                // If true, add i to sum
                sum += i;
            }
        }
        // Check if sum is greater than number
```



```

        if (sum > number) {
            System.out.println(number + " is an Abundant number.");
        }
        else {
            System.out.println(number + " is not an Abundant
number.");
        }
        // Close the scanner
        input.close();
    }
}

```

- Write a program **DayOfWeek** that takes a date as input and prints the day of the week that the date falls on. Your program should take three command-line arguments: m (month), d (day), and y (year). For m use 1 for January, 2 for February, and so forth. For output print 0 for Sunday, 1 for Monday, 2 for Tuesday, and so forth. Use the following formulas, for the Gregorian calendar (where / denotes integer division):

$$y_0 = y - (14 - m) / 12$$

$$x = y_0 + y_0/4 - y_0/100 + y_0/400$$

$$m_0 = m + 12 \times ((14 - m) / 12) - 2$$

$$d_0 = (d + x + 31m_0 / 12) \bmod 7$$

```

public class DayOfWeek {
    public static void main(String[] args) {

        // Take command-line arguments for month, day, and year
        int m = Integer.parseInt(args[0]);
        int d = Integer.parseInt(args[1]);
        int y = Integer.parseInt(args[2]);

        // Calculate y0, x, m0, and d0 using the provided formulas
        int y0 = y - (14 - m) / 12;
        int x = y0 + y0 / 4 - y0 / 100 + y0 / 400;
        int m0 = m + 12 * ((14 - m) / 12) - 2;
        int d0 = (d + x + 31 * m0 / 12) % 7;
    }
}

```

```
        // Print the day of the week (0 = Sunday, 1 = Monday, ..., 6
= Saturday)
        System.out.println(d0);
    }
}
```

6. Write a program to create a calculator using **switch...case**.

Hint =>

- Create two double variables named first and second and a String variable named op.
- Get input values for all variables.
- The input for the operator can only be one of the four values: "+", "-", "*" or "/".
- Run a for loop from i = 1 to i < number.
- Based on the input value of the op, perform specific operations using the **switch...case** statement and print the result.
- If op is +, perform addition between first and second; if it is -, perform subtraction and so on.
- If op is neither of those 4 values, print Invalid Operator.

```
import java.util.Scanner;
public class Calculator {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Get input values for the variables
        System.out.print("Enter the first number: ");
        double first = input.nextDouble();
        System.out.print("Enter the second number: ");
        double second = input.nextDouble();
        System.out.print("Enter an operator (+, -, *, /): ");
        String op = input.next();

        // Perform specific operations based on the input value of the
operator
        switch (op) {
            case "+":
                System.out.println("Result: " + (first + second));
                break;
```

```
        case "-":
            System.out.println("Result: " + (first - second));
            break;
        case "*":
            System.out.println("Result: " + (first * second));
            break;
        case "/":
            // Handle division by zero
            if (second != 0) {
                System.out.println("Result: " + (first /
second));
            }
            else {
                System.out.println("Division by zero is not
allowed.");
            }
            break;
        default:
            System.out.println("Invalid operator.");
            break;
    }
    // Close the scanner
    input.close();
}
}
```